CampusDesk: support across campus

01 Track classroom, lab and account incidents with a required campus location.

02 Classify requests as hardware, software, network, account or other.

03 Shared ResolveDesk foundation and AI assistance are credited explicitly.

From a lab incident to a resolution

01 Example: projector loses HDMI signal in Science block, Lab 301.

02 Capture requester, location, category, priority and assigned support team.

03 Move Open to In progress to Resolved; search by location or category.

04 Demonstration data is synthetic. The application has no authentication or email delivery.

Campus fields have their own contract

01 TicketForm and TicketTable collect and show category and location.

02 Pydantic requires location and validates the category enum; partial updates reject nulls.

03 Routes call service functions; SQLAlchemy stores the ticket in PostgreSQL.

04 Migration 0003 adds campus fields without deleting existing ticket history.

One shared cluster, separate campus data

Browser -> **React / Nginx** -> **FastAPI** -> **PostgreSQL**

GitHub Actions -> GHCR -> ArgoCD / Helm -> EKS

Prometheus <- /metrics Grafana <- Prometheus

01 Local Compose: browser 3002, API 8001, private PostgreSQL service.

02 Recorded EKS: namespace campusdesk, separate Secret and persistent database.

A tested image reaches the registry

01 31 backend tests and four frontend tests passed; the production frontend build passed.

02 Commit-SHA images passed the Trivy HIGH/CRITICAL gate before publication.

03 The same images passed Helm deployment and HTTP smoke checks in temporary Kind.

04 Final published baseline CI run 38906213468 passed at commit 1bd48967.

Quota recovery changed the network plan

01 The regional Elastic IP quota blocked the initial private-worker/NAT plan.

02 Approved recovery used public-subnet workers with no public inbound access and API /32.

03 Replacement workers served both applications; the old failed node group delayed removal.

04 Recovery apply was gracefully interrupted during deletion; it was not a clean successful apply.

The campus workload was verified

01 Recorded proof shows two API replicas, two frontend replicas and a completed migration Job.

02 PostgreSQL used its own bound 5 GiB volume in namespace campusdesk.

03 Readiness, liveness, resource limits and HPA were configured; metrics were available.

04 The cloud cluster has since been destroyed. Today use local UI and recorded cloud evidence.

GitOps was more than a YAML file

01 ArgoCD tracked the CampusDesk repository and Helm release values.

02 The recorded application reached Synced, Healthy and Succeeded.

03 Deployed images used the verified 9a7a184 commit tag; later docs commits kept that release.

04 Backend replica drift is ignored so ArgoCD does not fight the HPA.

Check actual metrics, not assumptions

01 Recorded Prometheus proof shows both CampusDesk API endpoints UP.

02 The campus dashboard contains real request-rate and p95-latency samples.

03 The error panel had no matching series; that was not presented as a measured zero.

04 Grafana authentication was refreshed after a monitoring pod restart.

Delivery finished with a cleanup audit

01 31 backend tests, four frontend tests, browser/mobile checks and PostgreSQL smoke passed.

02 A ten-slide deck and a narrated walkthrough document the delivery.

03 Shared Terraform destroy removed 59 resources; all 11 scoped cleanup checks passed.

04 The owned KMS key is PendingDeletion until 15 November 2026 under AWS rules.

05 Local demo remains available. Production would need authentication, backups and recovery planning.